ModelIt!

Student Activity Pack

Is Climate Change Reversible?

Modeling Carbon Cycle Dynamics and Climate Tipping Points

Grade 12L2 | NGSS HS-ESS3-5 | Stability and Change

STUDENT NAME _____ DATE _____

ModelIt! — Discovery Collective

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1 Pre-Assessment

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What Do You Know? Answer each question below to the best of your ability. There are no wrong answers — this helps your teacher understand what you already know!

1. **What do you think would happen to global temperatures if all fossil fuel burning stopped tomorrow? Would temperatures drop immediately, stay the same, or keep rising?**

2. **Can you explain what a feedback loop is? Can you give an example from any system you know about?**

3. **Where does the carbon go when you burn gasoline in a car? Trace the carbon atoms through the system.**

4. **Do you think climate change can be reversed? What would it take, and how long do you think it would require?**

2 Scoring Rubric

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Your teacher will use this rubric to assess your work. Read through each level to understand what is expected!

Skill	Expert (4)	Proficient (3)	Developing (2)	Beginning (1)
Identifying Components	EXPERT Correctly identifies and sorts ALL components as internal or external with detailed reasoning	PROFICIENT Correctly identifies and sorts MOST components with some reasoning	DEVELOPING Identifies components but has difficulty sorting or explaining	BEGINNING Needs significant help identifying components
Establishing Relationships	EXPERT Accurately describes ALL positive and negative relationships with clear evidence	PROFICIENT Accurately describes MOST relationships with some evidence	DEVELOPING Identifies some relationships but struggles with direction	BEGINNING Needs help understanding how components connect
Running Simulations	EXPERT Makes accurate predictions AND explains surprising results using evidence	PROFICIENT Makes reasonable predictions and records clear observations	DEVELOPING Runs simulations but predictions or observations are incomplete	BEGINNING Needs help running simulations or recording data
Explaining Results	EXPERT Uses multiple pieces of evidence to explain HOW and WHY systems interact	PROFICIENT Uses evidence to explain system interactions	DEVELOPING Gives partial explanations with limited evidence	BEGINNING Needs help connecting evidence to explanations

3 Component Analysis

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What is a Component?

A **component** is a part of a system that plays a specific role. Components can be **EXTERNAL** (come from outside the system) or **INTERNAL** (happen within the system).

Sort These Components

- Atmospheric CO2
- Fossil Fuel Emissions
- Ocean Carbon Absorption
- Terrestrial Carbon Storage
- Global Temperature Anomaly
- Ice-Albedo Feedback
- Permafrost Methane Release

EXTERNAL	INTERNAL

Explain your reasoning: Why did you sort each component the way you did?

4 Model Recording Page

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Draw your model showing all components and their relationships. Use the relationship key below to label your arrows!

+ **POSITIVE:** When A increases, B also increases

− **NEGATIVE:** When A increases, B decreases

Draw your model here

? Explain ONE relationship in your model. What happens when one component changes?

5 Simulation Observations

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Run each scenario in ModelIt! First, write your **prediction**. Then run the simulation and record what you **observe**.

1 Scenario 1: Business As Usual

Fossil Fuel Emissions: Current Rate | Ocean Absorption: Declining | Terrestrial Storage: Declining

MY PREDICTION

WHAT I OBSERVED

2 Scenario 2: Immediate Net Zero

Fossil Fuel Emissions: Zero | Ocean Absorption: Current | Terrestrial Storage: Current

MY PREDICTION

WHAT I OBSERVED

3 Scenario 3: Enhanced Sinks Strategy

Fossil Fuel Emissions: 50% Reduction | Ocean Absorption: Current | Terrestrial Storage: Increased 30%

MY PREDICTION

WHAT I OBSERVED

6 Research & Extend

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Real systems are more complex than our model! Here are additional components scientists study. Choose one and explain how it could connect to our model.

Possible New Components

Aerosol Masking Effect — The temporary cooling caused by sulfate aerosols from industrial pollution that reflect sunlight — paradoxically, reducing air pollution unmasks additional warming that was being hidden, meaning cleaner air initially increases temperatures

Deep Ocean Circulation — The global thermohaline conveyor belt that distributes heat and dissolved carbon through ocean basins over centuries — disruption of this circulation by freshwater from melting ice sheets could dramatically alter regional climates

Methane Clathrate Stability — Frozen methane deposits on the ocean floor and in permafrost that become unstable as temperatures rise — the potential release of this massive methane reservoir represents one of the most concerning climate tipping points

My Extension

A Which new component would you add to the model?

B How would it connect to the existing components? (Draw or describe)

C What evidence or reasoning supports your connection?

7 Reflection Questions

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The Big Question

If humanity stopped all carbon emissions tomorrow, would the climate return to normal — or have we already pushed Earth past points of no return that will reshape the planet for thousands of years regardless of what we do next? Use evidence from your simulation to support your answers.

1. **How does your model demonstrate that the relationship between emissions and temperature is not a simple linear cause-and-effect but involves delays, feedbacks, and thresholds?**

2. **Why do positive feedback loops make climate change harder to reverse than it was to start? Use specific examples from your model.**

3. **What does the concept of thermal inertia mean for policy decisions being made today? Why does delay matter so much?**

4. **How would you explain to someone who says \'it\'s too late to do anything\' that while some changes are irreversible, action still matters enormously?**

5. **What are the limitations of your seven-component model for representing the full complexity of the global climate system?**

8 STEM Challenge: Design a Regional Climate Resilience Plan

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YOUR ENGINEERING MISSION

Design a comprehensive climate action plan for a specific region that combines emission reduction, carbon sequestration, and adaptation strategies based on your model\'s predictions about feedback loops and tipping points.

THE SCENARIO

A coastal city of 500,000 people faces multiple climate threats: sea level rise from ice sheet melting, increased storm intensity from warmer ocean temperatures, reduced water supply from changing precipitation patterns, and heat waves that exceed historical records. The city council has commissioned your team to design a 30-year climate resilience plan that addresses both mitigation (reducing emissions) and adaptation (preparing for impacts that are already locked in). Your plan must be scientifically grounded in the carbon cycle dynamics and tipping point analysis from your model.

Engineering Design Thinking

1. WHAT SPECIFIC EMISSION REDUCTION TARGETS DOES YOUR PLAN SET, AND WHAT EVIDENCE FROM YOUR MODEL SUPPORTS THESE TARGETS?

2. HOW DOES YOUR PLAN ADDRESS THE IRREVERSIBLE CHANGES THAT ARE ALREADY LOCKED IN DUE TO THERMAL INERTIA AND FEEDBACK LOOPS?

3. WHAT NATURAL AND TECHNOLOGICAL CARBON SINKS DOES YOUR PLAN INCORPORATE, AND WHAT IS THEIR PROJECTED IMPACT?

4. HOW DOES YOUR PLAN PRIORITIZE ACTIONS THAT PREVENT TIPPING POINTS FROM BEING CROSSED?

My Design

DESIGN AREA — DRAW YOUR SOLUTION

N

! **Explain your design using scientific evidence from the simulation:**

Real Career Connection

Climate scientists and atmospheric researchers study Earth\'s climate systems, build predictive models, and analyze data from satellites, ice cores, and ocean sensors. They work at NASA, NOAA, universities, and private research firms, earning $70,000-$140,000/year. Climate policy analysts who translate scientific findings into actionable government and corporate strategies earn $65,000-$130,000/year. Carbon market analysts and sustainability consultants earn $75,000-$150,000/year in the rapidly growing green economy.